

RESUME

RAHUL SHAKYA

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OBJECTIVE:

To work with an organization where I can learn new skills and increase my abilities for the organization goal as well as myself.

An Enthusiastic **Embedded Systems, DSPs and Wireless Communication Engineer**, eagerly seeks an engaging opportunity in a dynamic tech environment. Eager to apply my diverse skill set and knowledge to drive innovation and contribute to the success to the team.

➤ PG Diploma in **Embedded Systems Design** from **C-DAC Pune** 26/08/2024 – 12/02/2025

Course Modules: -

- Embedded C Programing
- Microcontroller Programming and Interfacing
 - **Arm Cortex M**, RISC-V, **STM32F407**, WDT, SysTick, PWM, GPIO etc.
 - **Communication protocol: RS-232, UART, SPI, I2C, CAN, USB, Firmware**
- Embedded Operating Systems
- **Real-Time Operating Systems I.e. Free RTOS**
- Embedded Linux Device Drivers
- Data Structures and Algorithms
- Internet of Things (IoT)
 - LPWAN (LoRa, NB- IoT), **6LowPAN, REST, CoAP, MQTT, TCP, UDP**
- ❖ **Project:** Vehicle Automation System i.e Rainwater Drop Sensor, Buzzer Collision Detection, LED Intensity Controller Using CAN Protocol and PWM

WORK EXPERIENCE

Associate DSP Engineer

6/Nov/2023 – 28/March/2024

Constelli Signals Pvt Ltd, Hyderabad, TS.

❖ **Projects:**

- Audio Signal Processing, Signal Reconstruction, Noise Filtration, Spectrum Analyzer, DUC, DDC
- 16 QAM Implementation (Symbol mapping, Interpolation, Decimation, RRC Filter, PLL Detector, Matched Filter, Constellation Diagram, BER analysis)

Wireless Communication Engineer

Gigayasa Wireless, IITM Research Park, Chennai, TN

16/Jan/2023 – 30/Oct/2023

❖ Projects:

- Implemented Physical Layer algorithm: **Synchronization Signal Block (SSB), UE Measurement** such as SS-RSRP, CSI-RSRP, SS-RSRQ, CSI-RSRQ, CSI-SINR for 5G.
- Implemented 5G Modulations, Symbol Mapping, Pulse Shaping.
- Experience with **Software Defined Radio** using PySDR.
- Implemented Digital Modulation Techniques: ASK, FSK, PSK, QPSK, BPSK, Mary PSK, Mary ASK, Mary FSK, QAM with Symbol mapping and de-mapping using hard-decision and soft-decision i.e., log MAP, LLR.
- Implemented Communication Toolkit: PCM, PAM, PWM, PPM, DPCM, DM, ADM, PM, FM, AM, DSB-SC, SSB-SC for Transmitter and Receiver side.
- **Understanding of Physical Channel and Physical Signal** i.e. PBCH, PDCCH, PDSCH, PRACH, PUCCH, PUSCH, DMRS, CSI-RS, PT-RS, SRS
- **AREA OF INTEREST:** - Microcontrollers, OFDM (CP- OFDM, DFT-S-OFDM), massive MIMO, Beamforming, DSP algorithm (sampling, quantization, ADC, Channel coding, DAC, Modulations, Filters, FFT, Convolution etc.

Systems Engineer

Infosys Limited, Indore, MP

17/Jan/2022 – 12/Jan/2023

- Oracle, SQL/PLSQL

TRAINING

- Vocational Training in GSM, CDMA and WiMAX Technologies
BSNL, Bhopal, MP

Jun/2017 – Jul/2017

TECHNICAL SKILLS:

- Python [NumPy, SciPy, Matplotlib, Sphinx] with API documentation
- MATLAB
- LabVIEW
- C/C++
- Linux(ubuntu)

EDUCATIONAL QUALIFICATION:

- M.Tech in Electronics and Telecommunication Engineering (Specialization with **Communication Engineering and Signal Processing**) From **IIEST Shibpur**, Howrah/Kolkata with **8.43 CGPA** (2020-22)
- B.E. in **Electronics and communication Engineering** From **SATI, Vidisha** (RGPV University) with **7.01 CGPA** (2015-19).

ACADEMIC PROJECTS:

- **Free Space Optical Communication:** -
Channel Estimation reduction in bit error rate by using adaptive detection threshold with the help of tracking mean and variance using Kalman filter. Sept 2021 – Aug 2022(M.Tech)
- **Smart Laboratory Room Manager:** - Bidirectional Visitor Counter Jan 2019 – May2019(B.E)

ACCOMPLISHMENT:

- **Certifications**

Infosys certified Wireless LTE Specialist

Digital Signal Processing (DSP) From Ground Up™ in C (Udemy)

Microcontroller Embedded C Programming: Absolute Beginners (Udemy)

- **Competitions**

C-CAT-2024, Organized by C-DAC Pune (Qualified)

GATE-2020, Organized by IIT Delhi (Qualified)

LANGUAGES KNOWN:

- English and Hindi

Date:

Rahul Shakya